Final Project

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STAT 385 Final Project

Tetuan City Power Consumption Analysis

```r
#setwd("C:/Users/yasmi/OneDrive/Desktop/Final Project")
data <- read_csv("Tetuan City power consumption.csv")
```

```
## Rows: 52416 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): DateTime
## dbl (8): Temperature, Humidity, Wind Speed, general diffuse flows, diffuse f...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(data)
```

```
## Rows: 52,416
## Columns: 9
## $ DateTime                    <chr> "1/1/2017 0:00", "1/1/2017 0:10", "1/1/201~
## $ Temperature                 <dbl> 6.559, 6.414, 6.313, 6.121, 5.921, 5.853, ~
## $ Humidity                    <dbl> 73.8, 74.5, 74.5, 75.0, 75.7, 76.9, 77.7, ~
## $ `Wind Speed`                <dbl> 0.083, 0.083, 0.080, 0.083, 0.081, 0.081, ~
## $ `general diffuse flows`     <dbl> 0.051, 0.070, 0.062, 0.091, 0.048, 0.059, ~
## $ `diffuse flows`             <dbl> 0.119, 0.085, 0.100, 0.096, 0.085, 0.108, ~
## $ `Zone 1 Power Consumption`  <dbl> 34055.70, 29814.68, 29128.10, 28228.86, 27~
## $ `Zone 2  Power Consumption` <dbl> 16128.88, 19375.08, 19006.69, 18361.09, 17~
## $ `Zone 3  Power Consumption` <dbl> 20240.96, 20131.08, 19668.43, 18899.28, 18~
```

```r
summary(data)
```

```
##       DateTime      Temperature        Humidity       Wind Speed
##  Length   :52416   Min.   : 3.247   Min.   :11.34   Min.   :0.050
##  N.unique :52416   1st Qu.:14.410   1st Qu.:58.31   1st Qu.:0.078
##  N.blank  :    0   Median :18.780   Median :69.86   Median :0.086
##  Min.nchar:   13   Mean   :18.810   Mean   :68.26   Mean   :1.959
```

```
##  Max.nchar:   16   3rd Qu.:22.890   3rd Qu.:81.40   3rd Qu.:4.915
##                    Max.   :40.010   Max.   :94.80   Max.   :6.483
##  general diffuse flows diffuse flows     Zone 1 Power Consumption
##  Min.   :   0.004      Min.   :  0.011   Min.   :13896
##  1st Qu.:   0.062      1st Qu.:  0.122   1st Qu.:26311
##  Median :   5.035      Median :  4.456   Median :32266
##  Mean   : 182.697      Mean   : 75.028   Mean   :32345
##  3rd Qu.: 319.600      3rd Qu.:101.000   3rd Qu.:37309
##  Max.   :1163.000      Max.   :936.000   Max.   :52204
##  Zone 2  Power Consumption Zone 3  Power Consumption
##  Min.   : 8560             Min.   : 5935
##  1st Qu.:16981             1st Qu.:13129
##  Median :20823             Median :16415
##  Mean   :21043             Mean   :17835
##  3rd Qu.:24714             3rd Qu.:21624
##  Max.   :37409             Max.   :47598
```

```r
sum(duplicated(data))   # No duplicates
```

```
## [1] 0
```

```r
# The dataset contains multiple timestamp formats:
# - "1/1/2017 0:00"
# - "01/01/2017 00:00"
# - "2017-01-01 00:00:00"
# parse_date_time() handles all of them.

data <- data %>%
  mutate(DateTime = parse_date_time(
    DateTime,
    orders = c("d/m/Y H:M",
               "d/m/Y H:M:S",
               "Y-m-d H:M:S",
               "m/d/Y H:M",
               "m/d/Y H:M:S")
  ))

# Verify successful parsing
sum(is.na(data$DateTime))   # Should be 0
```

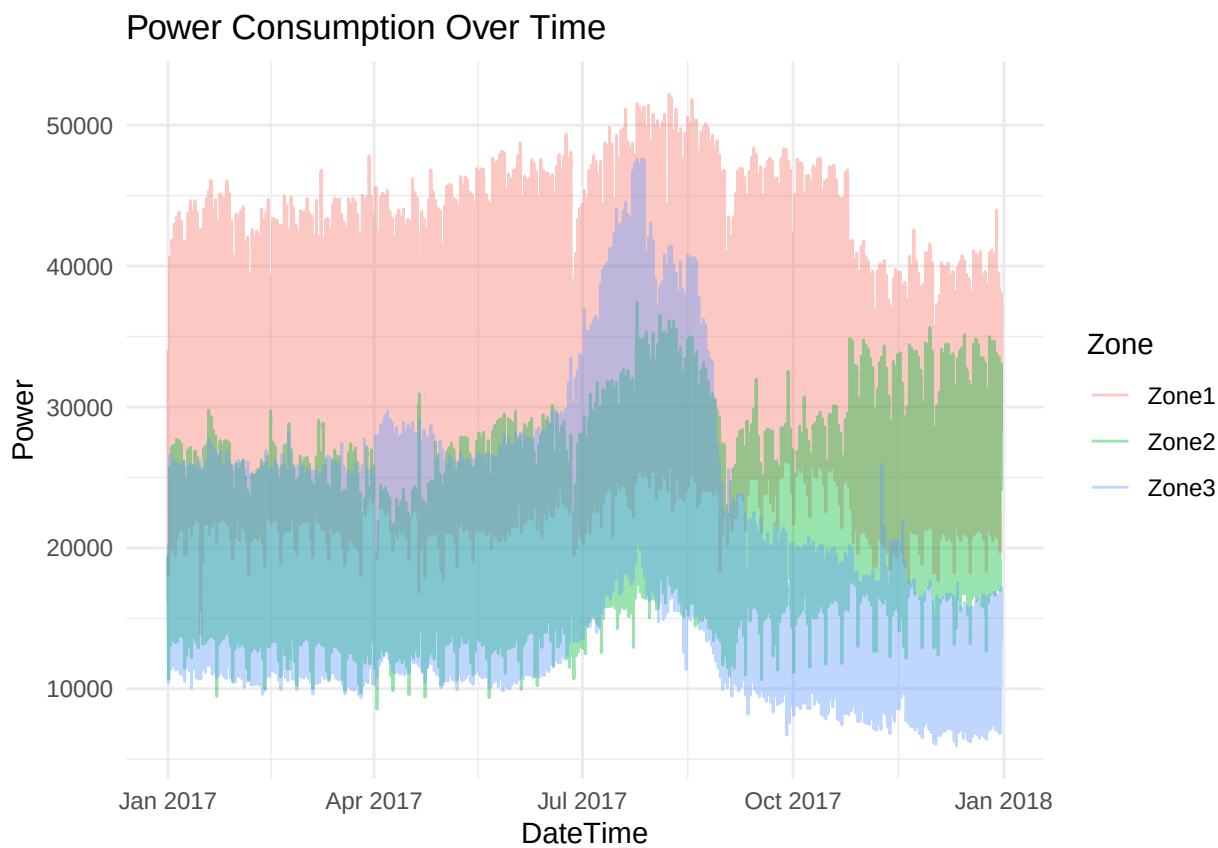
```
## [1] 0
```

```r
range(data$DateTime)        # Should show full 2017 range
```

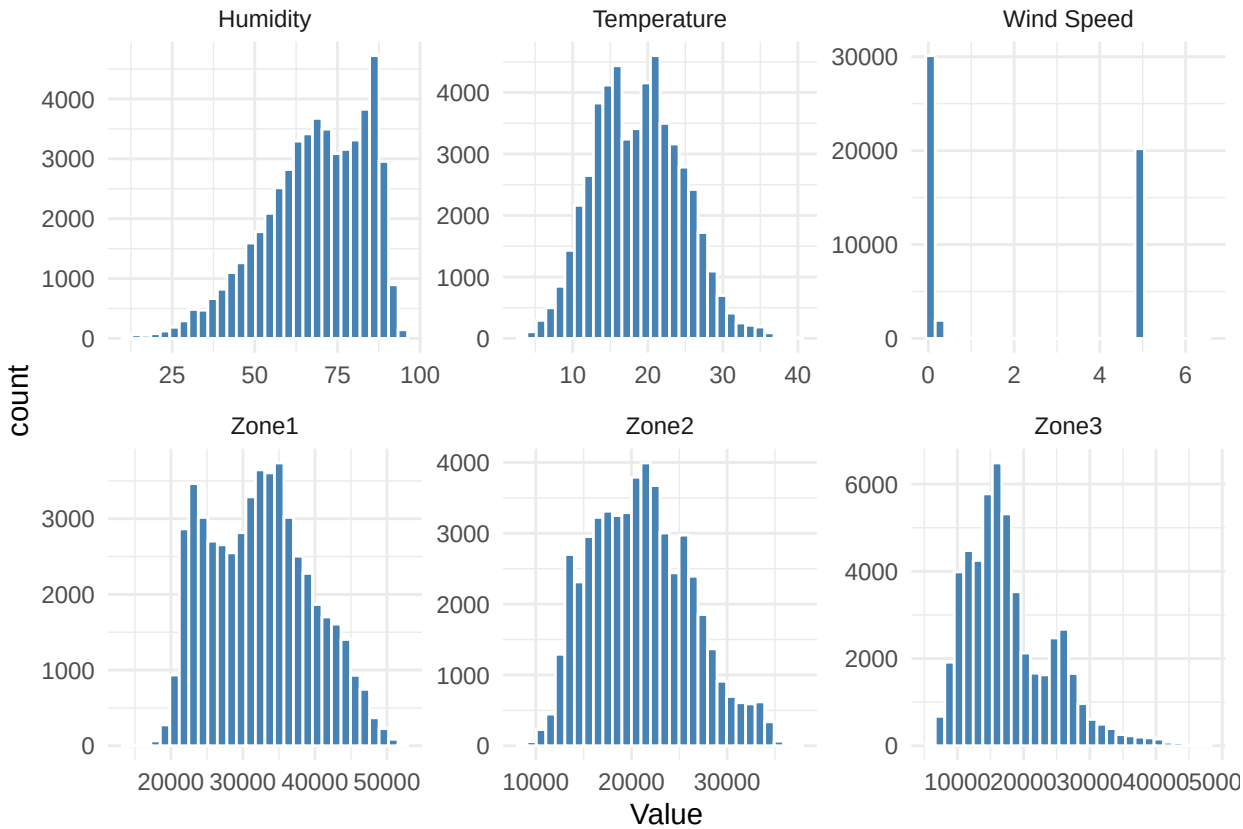
```
## [1] "2017-01-01 00:00:00 UTC" "2017-12-30 23:50:00 UTC"
```

```r
data <- data %>%
  rename(
    Zone1 = `Zone 1 Power Consumption`,
    Zone2 = `Zone 2  Power Consumption`,   # two spaces in original
    Zone3 = `Zone 3  Power Consumption`    # two spaces in original
  )
```

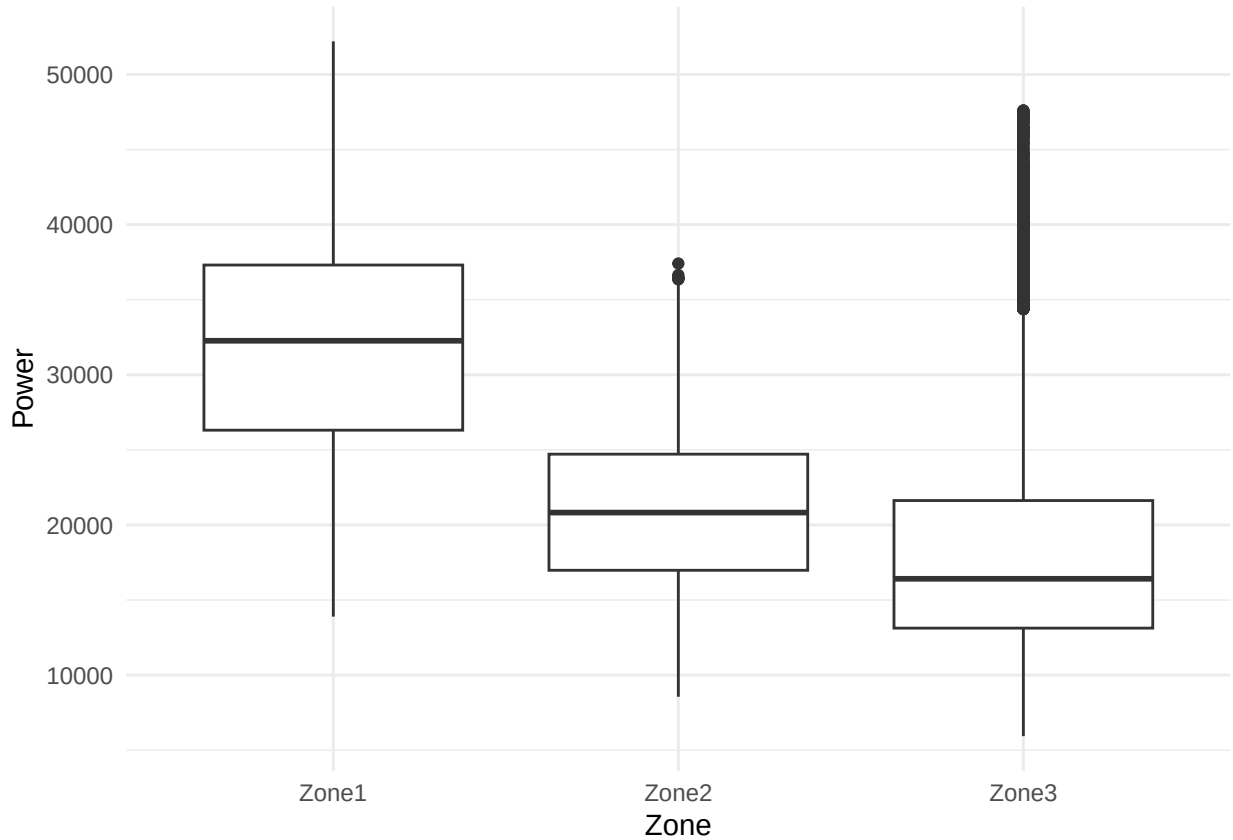
```r
## 4a. Time Series Plot for All Zones
data %>%
  select(DateTime, Zone1, Zone2, Zone3) %>%
  pivot_longer(-DateTime, names_to = "Zone", values_to = "Power") %>%
  ggplot(aes(DateTime, Power, color = Zone)) +
  geom_line(alpha = 0.4) +
  labs(title = "Power Consumption Over Time", y = "Power") +
  theme_minimal()
```




```r
## 4b. Histograms for All Variables
data %>%
  select(Temperature, Humidity, `Wind Speed`,
         Zone1, Zone2, Zone3) %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(Value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal()
```

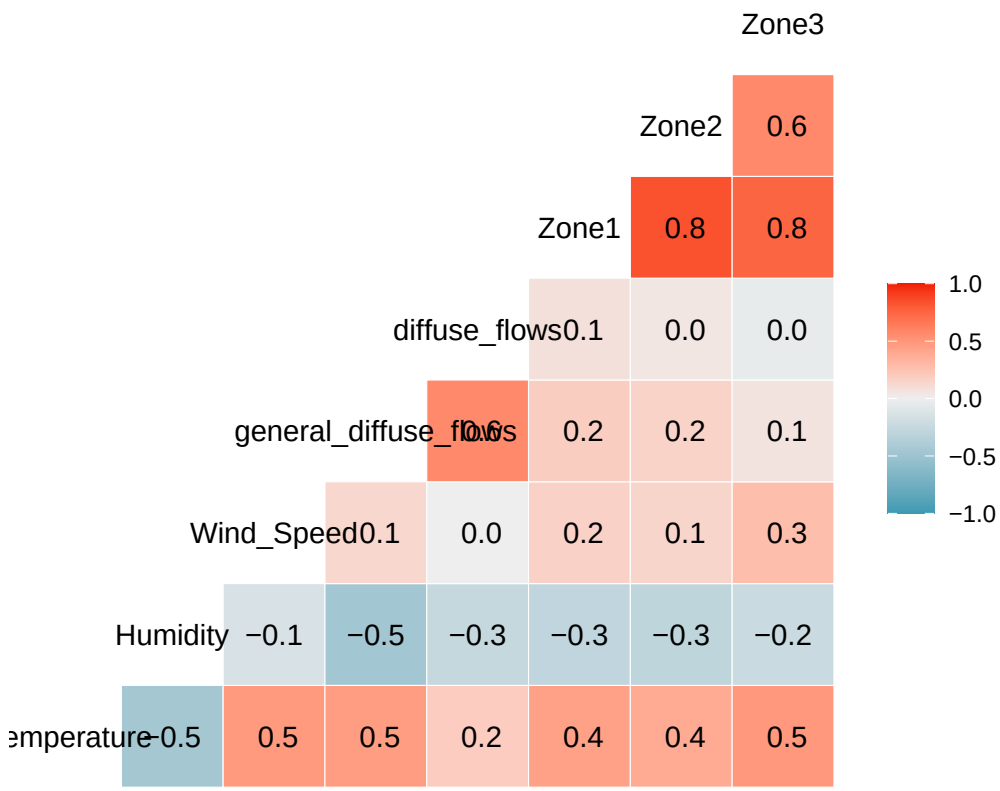



```r
## 4c. Boxplots for Zone Power Consumption
data %>%
  select(Zone1, Zone2, Zone3) %>%
  pivot_longer(everything(), names_to = "Zone", values_to = "Power") %>%
  ggplot(aes(Zone, Power)) +
  geom_boxplot() +
  theme_minimal()
```




```r
## 4d. Correlation Matrix
data_corr <- data %>%
  select(Temperature, Humidity, `Wind Speed`,
         `general diffuse flows`, `diffuse flows`,
         Zone1, Zone2, Zone3)

ggcorr(data_corr, label = TRUE)
```




```r
data <- data %>%
  mutate(
    hour  = hour(DateTime),
    dow   = wday(DateTime, label = TRUE, week_start = 1),
    month = month(DateTime, label = TRUE)
  )
```

```r
num_vars <- c("Temperature", "Humidity", "Wind Speed",
              "general diffuse flows", "diffuse flows")

data_scaled <- data %>%
  mutate(across(all_of(num_vars), scale))
```

```r
n <- nrow(data_scaled)
train_end <- floor(0.8 * n)

train <- data_scaled[1:train_end, ]
test  <- data_scaled[(train_end + 1):n, ]
```